14.3.2 Enzymes Involved in DNA Replication

DNA replication in *E.coli* requires many enzymes and proteins, each performing a specific task. The entire complex is called the DNA replicase system or replisome. The major enzymes and proteins involved with their functions are tabulated in Table 14.2.

Table 14.2: Enzymes involved in DNA replication

Enzyme	Function
Helicase	Unwinds DNA
DNA gyrase	Relieves stress created by unwinding
SSB protein	Binds to single stranded DNA and stabilises it
Primase	Synthesis of RNA primer
DNA pol I	Excision of primers and filling of gaps with nucleotides
DNA pol III	New strand elongation
DNA ligase	Joins the nick

Infobits

- First in vitro synthesis of DNA with a template was carried out by Kornberg in 1959.
- First in vitro synthesis of DNA without template was carried out by H.G. Khorana in 1965.

14.3.3 Events in DNA Replication in *E.coli*

1. Initiation
2. Elongation
3. Termination

DNA replication is depicted in Figure 14.14.

Initiation

- DNA replication is initiated at replication origin known as oriC (245 base pairs in *E.coli*).
- Dna A protein molecules bind to the origin of replication.
- Helicase (DnaB) denatures the DNA helix by breaking the hydrogen bonds between base pairs.
- Many molecules of SSB (single stranded binding) proteins bind cooperatively to single stranded DNA, stabilizing the separated strands and preventing renaturation.
- Gyrase (topoisomerase) releases the topological stress produced by helicase.
- RNA primers are synthesised by primase.

The separated polynucleotide strands are used as templates for synthesis of complementary strands. The area of the DNA opened by helicase for DNA synthesis is referred to as the replication fork. At the replication fork there are four strands of DNA, two are conserved and two are newly synthesised. Replication may occur in either a unidirectional or bidirectional manner (Figure 14.13) from each origin. Bidirectional replication can be explained as two replication forks moving in opposite directions around the circular chromosome. Both forks move along the double helix away from the origin of replication in opposite directions and around the circular chromosome.

Elongation

- DNA synthesis proceeds in a 5′→3′ direction(read as 5 prime to 3 prime).
- One strand is synthesised continuously and is known as leading strand.

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